# Morteza Poudat

Application Consultant | Python & Web Development | Data Integration

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## PROFILE

Application Consultant and Python developer with an engineering background (M.Sc. Computational Mechanics). At Matplus GmbH since 2017, taking customer projects from requirements workshops to delivered plugins, web applications, data migrations and data models.

## KEY SKILLS

- **Customer consulting & project management:** requirements workshops, solution design, implementation project management, customer meetings, cost proposals, customer support from issue to delivered feature
- **Web development:** web applications, business and portfolio websites, online booking systems, bilingual (DE/EN) sites; HTML, CSS, JavaScript, React, Next.js, TypeScript, Django
- **Data extraction, analysis & modelling:** data extraction from PDFs, scans, Excel, XML, JSON and web sources; data cleaning, validation and import; data modelling and migration; Python (Pandas), SQL, PostgreSQL, MongoDB
- **Automation & AI-assisted development:** AI coding agents (Cursor), Python automation, API integration, internal tools; Git/GitLab, Docker, Linux
- **Engineering:** FEM / computational mechanics (Abaqus, ANSYS APDL), materials data (steel, copper, aluminium alloys), Matlab, AutoCAD

## PROFESSIONAL EXPERIENCE

**Application Consultant & Python / Data Engineer** Jun 2017 – Present

*Matplus GmbH, Wuppertal · Application Consultant since Oct 2017*

- **Customer implementation projects:** Runs several customer implementation projects in parallel – consulting, requirements workshops, plugin design, data migration, weekly customer meetings and project management through to delivery.
- **Web application development:** Ongoing time tracking and staff-planning web app: gathered requirements in a customer meeting, wrote the full specification and implemented it AI-assisted (Cursor, coding agents) – organisational levels, role-based permissions, skill matrix and staff planning; Python, Git/GitLab.
- **Data migration:** Migrated a large energy-technology customer's legacy Microsoft Access database (structural and materials data) to MongoDB with Python importers, including matching legacy test numbers to master materials.
- **Reference database integration:** Integrates industry reference data into the company's materials data platform, recurring with each new edition: MMPDS (Excel editions), Stahldat (SQL database via KNIME ETL), Copper Key (APIs and PDFs), aluminium handbook data, FKM guideline, ESDU, NCAMP, CCT (ZTU) diagrams and metallography datasets. PDF extraction with PyMuPDF, OpenCV table/grid detection and OCR (Tesseract); every pipeline documented and re-runnable in GitLab.
- **Data modelling:** Designed the data model for a unified materials catalogue (steel, copper, aluminium) in MongoDB and defined the standard ETL process for all material imports.
- **Proposals & presales:** Prepared the proposal and cost estimate for a multi-phase materials-data platform for an aerospace research customer.
- **Conference speaker:** Talk on copper materials databases at the Copper Alloys conference, Düsseldorf.

**Civil Engineering Intern** 2009 – 2010

*Iran*

## SELECTED PROJECTS

- **Booking website for a Düsseldorf barbershop** — bilingual (DE/EN) mobile-first online booking with confirmation e-mails and calendar file, plus an admin area for bookings, customers, services and opening hours; Next.js, TypeScript, PostgreSQL.

- **Portfolio website for a Düsseldorf artist** — bilingual (DE/EN) one-page style portfolio presenting paintings and studio projects; HTML, CSS, JavaScript. (Website)
- **Personal portfolio website** — bilingual, fast and accessible site presenting freelance services and projects; Next.js, Tailwind. (Portfolio)

## EDUCATION

**M.Sc. Computational Mechanics** 2016 – 2021

*University of Duisburg-Essen, Germany · Grade 1.9*

- Thesis: Knowledge base for the selection and application of aluminium alloys.

**Materials Science & Applied Mechanics (semesters)** 2015 – 2016

*University of Duisburg-Essen, Germany*

**M.Sc. and B.Sc. Civil Engineering**

*Shahid Chamran University of Ahvaz, Iran*

## CERTIFICATES

- Real Python – Pandas (2021–2022)
- Sololearn – Python for Data Science; Intermediate Python (2021)
- MongoDB – MongoDB Basics (2018)
- MITx 6.00.1x – Introduction to Computer Science and Programming Using Python (2017)

## LANGUAGES

Persian – native | German – full professional proficiency | English – fluent | Arabic, Spanish – elementary